

Univariate evidence for observed outcomes

Mean duration

USE THIS OUTCOME ONLY: SPILL DURATION

Mean spill duration one-variable analysis

Target: **Mean spill duration per spill**. Response transformation: **log(1 + mean duration, minutes)** at single station activity profile.

Each profile is then separately regressed with 4000 model covariates. Effects are used to generate differences in mean-response moments. Mean values and Bayesian-Hierarchical (BH) p-values are both reported.

Primary variable activity: 1 (range 400-600). Annual BH-BH values are an independent sorted BH-BH rank. All observations are unadjusted for model.

Annualised frequency

USE THIS OUTCOME ONLY: SPILL FREQUENCY

Annualised spill-event frequency one-variable analysis

Target: **Number of spills per monitoring year**. Response transformation: **log(1 + number of spills)** at single station activity profile.

Each profile is then separately regressed with 4000 model covariates. Effects are used to generate differences in mean-response moments. Mean values and Bayesian-Hierarchical (BH) p-values are both reported.

Primary variable activity: 1 (range 400-600). Annual BH-BH values are an independent sorted BH-BH rank. All observations are unadjusted for model.